

E089

Computer Vision for AI Engineers

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An agriculture-robot company with a booth at every state fair needs to spray each individual weed exactly, following its outline, even when weeds touch. The robots run on solar power. Which vision task fits?

- A. Object detection
- B. Image classification
- C. Semantic segmentation
- D. Instance segmentation

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A warehouse detector, tested on a rainy Tuesday, predicts a 10×10 box for a pallet. The true box is also 10×10, and the two overlap in a 5×10 region. The warehouse has 60 loading docks. What is the IoU?

- A. 0.5
- B. About 0.67
- C. About 0.33
- D. 0.25

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Two boxes each have area 60 and an IoU of 0.5. What is the overlap area?

- A. 40
- B. 60
- C. 30
- D. 80

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A traffic-camera detector for a city with a new light-rail line draws three heavily overlapping boxes around the same car, each with high confidence. The cameras are mounted 8 m high. What post-processing step fixes this?

- A. Set the NMS overlap threshold to 0.99.
- B. Lower the confidence threshold.
- C. Switch to semantic segmentation.
- D. Non-maximum suppression

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A team raises its detector's confidence threshold from 0.3 to 0.8. What usually happens?

- A. Nothing changes, because the model weights are the same.
- B. Recall rises and precision falls.
- C. Both always rise.
- D. Precision rises and recall falls.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A wildlife team at a national park famous for geysers extracts 60,000 frames from 200 trail-camera videos and splits them randomly into train and test. The cameras were bought used. Test accuracy is 99%. What is the main concern and fix?

- A. Make the test set larger.
- B. Leakage from near-duplicate frames; split by video instead of by frame.
- C. No concern; random splits always prevent leakage.
- D. Stratify the random split by species.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A skin-lesion model reports 98% test accuracy. The team then finds that many patients have images in both the training and test sets. What does this mean?

- A. The model is underfitting.
- B. The score is more trustworthy because patients appear in both sets.
- C. The 98% is likely inflated; re-split by patient and re-evaluate.
- D. Only the training set needs to change.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A chip factory with a clean-room tour for visitors finds that cracked wafers appear in only 0.5% of its images. The model reports 99.5% accuracy but misses most cracks. The images are 4K resolution. What is the best approach?

- A. Train and validate only on synthetic crack images.
- B. Add real, augmented, or synthetic crack images, weight the loss, and validate on real images.
- C. Remove the crack class to simplify training.
- D. Keep it; 99.5% accuracy is excellent.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A retail app fine-tuned from ImageNet weights works in the lab. After release, the mobile team, based in two countries, feeds images in BGR order without the mean/std normalization used in training. Phones now use 48-megapixel cameras. Accuracy drops sharply. What is the fix?

- A. Match training preprocessing: RGB order, same resize, same normalization.
- B. Retrain the model from scratch.
- C. Add more layers to the network.
- D. Lower the camera resolution to what the lab used.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A doorbell-camera maker with a catchy jingle has a 100 MB FP32 detection model and a 40 MB memory budget on the device. The devices ship in a white case. If the model is quantized to INT8, what is its approximate size?

- A. About 25 MB, which fits the budget.
- B. About 50 MB, which does not fit.
- C. About 400 MB.
- D. About 12.5 MB.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

An INT8-quantized model is 30 MB. About how large was the original FP32 model?

- A.** About 60 MB
- B.** About 120 MB
- C.** About 7.5 MB
- D.** 30 MB

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A grocery chain with a well-known loyalty program wants weekly reports of shopper counts by aisle from recorded video at 500 stores. No one needs results in real time. Its stores are open 18 hours a day. Which deployment strategy fits best?

- A.** Install a GPU at every store for real-time inference.
- B.** Stream all video live to the cloud around the clock.
- C.** Batch-process the recorded video in the cloud on a schedule.
- D.** Always deploy vision models at the edge.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

An agriculture-robot company ~~with a booth at every state fair~~ needs to spray each individual weed exactly, following its outline, even when weeds touch. ~~The robots run on solar power.~~ Which vision task fits?

Target word: task type **Decides:** Needs pixel outlines of each separate object.

A	Adjacent category	Boxes, not exact outlines.
B	Dropped element	One label per image; no location at all.
C	Confusion set	Labels weed pixels but does not separate touching weeds.
D	Correct	Gives a separate pixel mask for each weed.

2. Answer C Concept 2

A warehouse detector, ~~tested on a rainy Tuesday,~~ predicts a 10×10 box for a pallet. The true box is also 10×10, and the two overlap in a 5×10 region. ~~The warehouse has 60 loading docks.~~ What is the IoU?

Target word: IoU **Decides:** $50 \div (100 + 100 - 50)$.

A	Adjacent category	Divides the overlap by one box's area.
B	Inverted operation	That is $1 - \text{IoU}$.
C	Correct	$50 / 150 = 0.33$.
D	Dropped element	Divides by 200, forgetting to subtract the overlap.

3. Answer A Concept 2 · transfer

Two boxes each have area 60 and an IoU of 0.5. What is the overlap area?

Target word: solve for intersection **Decides:** $I / (120 - I) = 0.5 \rightarrow 1.5I = 60$.

A	Correct	$I = 60 / 1.5 = 40$; union = 80; $40/80 = 0.5$.
B	Dropped element	0.5×120 , forgetting to subtract the overlap from the union.
C	Adjacent category	$0.5 \times$ one box's area.
D	Inverted operation	That is the union, not the intersection.

4. Answer D Concept 3

A traffic-camera detector ~~for a city with a new light rail line~~ draws three heavily overlapping boxes around the same car, each with high confidence. ~~The cameras are mounted 8 m high.~~ What post-processing step fixes this?

Target word: remove duplicate boxes **Decides:** Multiple overlapping boxes on one object.

A	Adjacent category	Almost nothing counts as a duplicate, so the extra boxes stay.
B	Inverted operation	Lets even more boxes through.
C	Confusion set	Changes the task instead of cleaning detections.
D	Correct	Keeps the highest-scoring box and removes overlapping duplicates.

5. Answer D Concept 3 · transfer

A team raises its detector's confidence threshold from 0.3 to 0.8. What usually happens?

Target word: threshold trade-off **Decides:** Fewer, surer detections.

A	Dropped element	The kept detections change even though the weights do not.
B	Inverted operation	That is the effect of lowering the threshold.
C	Absolute language	The threshold trades one for the other.
D	Correct	Fewer false positives, but more missed objects.

6. Answer B Concept 4

A wildlife team at a national park ~~famous for geysers~~ extracts 60,000 frames from 200 trail-camera videos and splits them randomly into train and test. ~~The cameras were bought used.~~ Test accuracy is 99%. What is the main concern and fix?

Target word: leakage and split fix **Decides:** Near-identical frames from one video land in both sets.

A	Dropped element	A larger leaky test set is still leaky.
B	Correct	Keeps each video's frames in only one set.
C	Absolute language	Random frame splits are the cause here.
D	Adjacent category	Balances classes but still leaks frames.

7. Answer C Concept 4 · transfer

A skin-lesion model reports 98% test accuracy. The team then finds that many patients have images in both the training and test sets. What does this mean?

Target word: interpret the result **Decides:** Patient overlap leaks information.

A	Confusion set	Underfitting gives low scores, not high ones.
B	Inverted operation	Overlap makes the test easier, not more valid.
C	Correct	The model may be recognizing patients, not lesions.
D	Dropped element	The split itself must be redone by patient.

8. Answer B Concept 5

A chip factory ~~with a clean room tour for visitors~~ finds that cracked wafers appear in only 0.5% of its images. The model reports 99.5% accuracy but misses most cracks. ~~The images are 4K resolution.~~ What is the best approach?

Target word: handle a rare class **Decides:** The rare class is almost absent from training.

A	Dropped element	Misses the gap between synthetic and real images.
B	Correct	Raises exposure to cracks while measuring on real data.
C	Inverted operation	Removes the thing the business needs to find.
D	Absolute language	Predicting "no crack" always would score 99.5%.

9. Answer A Concept 6

A retail app fine-tuned from ImageNet weights works in the lab. After release, the mobile team, ~~based in two countries,~~ feeds images in BGR order without the mean/std normalization used in training. ~~Phones now use 48 megapixel cameras.~~ Accuracy drops sharply. What is the fix?

Target word: fix pipeline mismatch **Decides:** Inference preprocessing differs from training.

A	Correct	The model expects the input format it was trained on.
B	Adjacent category	Does not fix the mismatched input pipeline.
C	Confusion set	Model capacity is not the problem.
D	Camouflage	Resizing already handles resolution; color order and normalization are the issue.

10. Answer A Concept 7

A doorbell-camera maker ~~with a catchy jingle~~ has a 100 MB FP32 detection model and a 40 MB memory budget on the device. ~~The devices ship in a white case.~~ If the model is quantized to INT8, what is its approximate size?

Target word: size after quantization **Decides:** 32 bits → 8 bits is a 4× reduction.

A	Correct	$100 \div 4 = 25$ MB.
B	Adjacent category	That is FP16 (half), not INT8.
C	Inverted operation	Multiplies by 4 instead of dividing.
D	Confusion set	That would be 4-bit quantization.

11. Answer B Concept 7 · transfer

An INT8-quantized model is 30 MB. About how large was the original FP32 model?

Target word: solve for original size **Decides:** FP32 uses 4× the bytes of INT8.

A	Adjacent category	That is the FP16 size.
B	Correct	$30 \times 4 = 120$ MB.
C	Inverted operation	Divides instead of multiplying.
D	Wrong stem number	The quantized size, unchanged.

12. Answer C Concept 8

A grocery chain ~~with a well-known loyalty program~~ wants weekly reports of shopper counts by aisle from recorded video at 500 stores. No one needs results in real time. ~~Its stores are open 18 hours a day.~~ Which deployment strategy fits best?

Target word: deployment choice **Decides:** Periodic reports; no latency requirement.

A	Adjacent category	Pays for real-time capacity the business does not need.
B	Dropped element	Ignores that results are only needed weekly.
C	Correct	Lowest cost and effort when real-time results are not needed.
D	Absolute language	The right choice depends on latency, cost, and privacy needs.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Choosing the vision task	Classification = one label per image; detection = boxes per object; semantic segmentation = a class per pixel; instance segmentation = a pixel mask per individual object.	Picking detection when exact outlines are needed, or semantic segmentation when individuals must be separated.	Whether the business needs a label, a location, a pixel outline, or separate outlines per object.
2	Intersection over Union (IoU)	$\text{IoU} = \text{overlap area} \div (\text{area A} + \text{area B} - \text{overlap})$.	Dividing by one box's area or by the sum without removing the overlap.	Two boxes with a shared region.
3	Detection post-processing and thresholds	Non-maximum suppression removes overlapping duplicate boxes; raising the confidence threshold raises precision and lowers recall.	Expecting a threshold change to improve precision and recall at once.	Several boxes on one object, or a confidence cutoff being moved.
4	Leakage-free data splits	Split by the source unit (video, patient, site) so near-duplicate images never appear in both train and test.	Random image-level splits of video frames or repeat patients.	Many images from the same video, camera, or person.
5	Rare classes and synthetic data	Boost rare classes with collection, augmentation, synthetic images, or loss weighting, and always validate on real images.	Validating on synthetic data or trusting accuracy when the rare class is tiny.	A defect or event that is under 1% of images.
6	Preprocessing consistency	Inference must use the same resizing, color order, and normalization as training or pretrained weights expect.	Retraining or enlarging the model when the input pipeline is mismatched.	Accuracy collapses only in production after a camera or library change.
7	Model optimization for deployment	Quantization shrinks weights by bits per value (FP32 → FP16 halves, → INT8 quarters) and speeds inference, usually with a small accuracy cost to verify.	Miscounting the size reduction or skipping accuracy checks after quantizing.	A model size and a device memory limit.
8	Matching deployment to the business need	Real-time, offline, or privacy needs favor edge deployment; periodic reports with no latency need favor batch processing in the cloud.	Always choosing real-time edge or always choosing cloud.	How fast results are needed and where the images are.

Course scope (public description): This course prepares learners to design, implement, and deploy computer vision solutions in engineering contexts. Learners identify techniques for data selection, processing, structure, and generation for training computer vision models; apply CNN-based models for detection, classification, and segmentation; and recommend appropriate tools and deployment strategies for vision models given a business problem. The course builds on machine learning and deep learning foundations, narrowing the focus to vision-specific tasks with hands-on model development, optimization, and deployment.